Regression and Sparsity

Regression Models

$$Y_{n\times 1} = X_{n\times p} B_{p\times 1} + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I)$$

$$p(B) \propto \text{const.} \qquad P(B \mid Y)?$$

$$P(z \mid \mu, \Sigma) \propto |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(z-\mu)^T \Sigma^{-1} (z-\mu)\right)$$

$z_{p\times 1} \sim N_p(\mu, \Sigma)$

$$\propto |\Sigma|^{-1/2} \exp\left(-\frac{1}{2} \operatorname{tr}\left(\Sigma^{-1}(z-\mu)(z-\mu)^T\right)\right)$$

$$\boxed{(z-\mu)^T \Sigma^{-1} (z-\mu) = \operatorname{tr}\left((z-\mu)^T \Sigma^{-1} (z-\mu)\right)}$$

$$\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$$

$$P(B \mid Y, X) \propto P(Y \mid B, X, \sigma^2)$$

$$\propto \exp\left(-\frac{1}{2}(y - XB)^T(\sigma^2 I)^{-1}(y - XB)\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}\,\mathrm{tr}\left[y^T y - 2B^T X^T y + B^T X^T X B\right]\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}\,\mathrm{tr}\left(X^T X\left(B^T B - 2\overbrace{\boxed{(X^T X)^{-1} X^T y}}^{\hat{B}_{MLE}}\, B\right)\right)\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}\,\mathrm{tr}\left(X^T X\left(B^T B - 2\hat{B} B\right)\right)\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}\,\mathrm{tr}\left(X^T X\,(B - \hat{B})^T(B - \hat{B}) - \cancel{\hat{B}^T \hat{B}}\right)\right)$$

$$\propto \exp\left(-\frac{1}{2}\,\mathrm{tr}\left((B - \hat{B})\left(\frac{X^T X}{\sigma^2}\right)(B - \hat{B})^T\right)\right)$$

$$p(B \mid y, X, \sigma^2) \sim N_p\left(\hat{B}_{MLE},\ \sigma^2 \underbrace{(X^T X)^{-1}}_{\text{must be invertible}}\right)$$

If $p > n$, $\underline{X^T X}$ is not full rank, hence not invertible. Posterior is not integrable.

But if, $p(\beta) \sim N(0, \sigma^2 \Lambda)$, then

$P(\beta | y, X, \sigma^2)$ is proper.

$$P(\beta | y, X, \sigma^2) \propto L(\beta) P(\beta) \propto$$

$$\exp\left(-\frac{1}{2\sigma^2} tr\left(X^T X \beta^T \beta - 2X^T y \beta\right)\right) \exp\left(-\frac{1}{2\sigma^2} tr\left(\beta \Lambda^{-1} \beta^T\right)\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2} tr\left(X^T X \beta^T \beta - 2X^T y \beta + \Lambda^{-1} \beta^T \beta\right)\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2} tr\left((X^T X + \Lambda^{-1})\left(\beta^T \beta - \underbrace{(X^T X + \Lambda^{-1})^{-1} X^T y \beta}_{\hat{\beta}_{Ridge}}\right)\right)\right)$$

$$P(\beta | \sigma^2, X, y) \sim N_p\left(\hat{\beta}_{Ridge}, \sigma^2 (X^T X + \Lambda^{-1})^{-1}\right)$$

$$\hat{\beta}_{Ridge} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda\|\beta\|^2$$

$$= (X^TX + \Lambda^{-1})^{-1} X^T y$$

$$= \underbrace{(X^TX + \Lambda^{-1})^{-1}(X^TX)}_{(I - K)} \underbrace{(X^TX)^{-1} X^T y}_{\hat{\beta}_{MLE}}$$

(Shrinkage Factor)

Let $\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_p \end{pmatrix}$, and X's are orthogonal, then

$$(I - K) = \begin{pmatrix} \ddots & & 0 \\ & \frac{\sum x_i^2}{\sum x_i^2 + 1/\lambda_i} & \\ 0 & & \ddots \end{pmatrix}$$

X^TX is diagonal.

$$E[B_j \mid X, y, \sigma^2] = (1 - K_j)\, \hat{B}_{OLS,j}$$

$$K_j = \frac{1}{1 + n s_j \lambda_j}$$

where $s_j = Var(x_j)$

Reminder: Scale Mixture Models

$Z|C \sim N(0, C)$

$C \sim$ Exponential

$Z \sim$ Laplace (heavier tails)

If $C \sim$ Inv-Gam.

$Z \sim$ t-distn.

⋮

Sparsity

- Lasso?
- "Spike and Slab"
- Horseshoe Priors

$$\min \|y - X\beta\|^2 + \lambda\|\beta\|_1$$

$$\beta_j \overset{iid}{\sim} \text{Laplace}, \quad P(\beta) \propto \exp(\lambda\|\beta\|_1)$$

$$L(\beta)P(\beta) \propto \exp(\|y - X\beta\|_2^2 + \lambda\|\beta\|_1)$$

MAP w/ Laplace priors is LASSO.

But, the posterior distn of β is dense in the coefs. almost everywhere!

Sparsifying Bayesian Priors

$$E[\beta_j \mid y, X, \sigma^2] = (1 - \kappa_j)\hat{\beta}_{OLS},$$

$$\kappa_j = \frac{1}{1 + n s_j \lambda_j}$$ Shrinkage factor.

Put a prior on λ_j, and integrate over it to get scale-mixture.

Study implied distn of κ_j.

Lasso and Laplace Priors

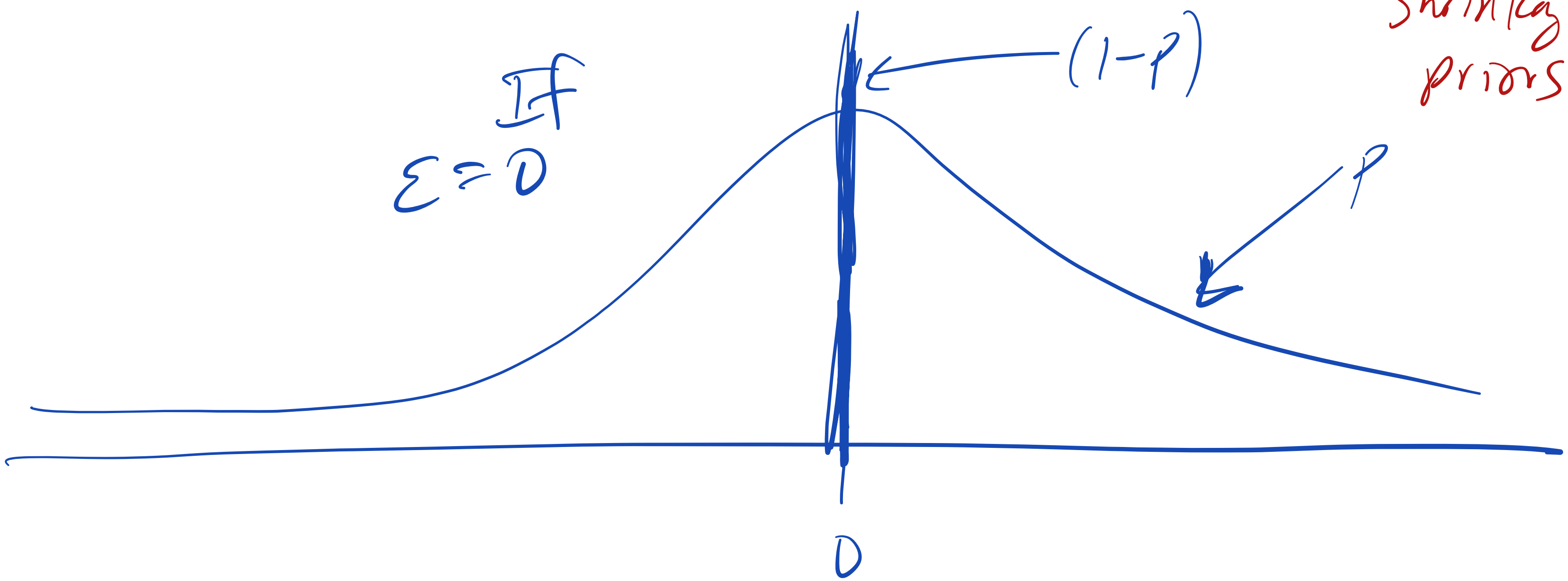
Spike and Slab

$B_j \mid \lambda_j, c \sim N(0, c^2 \lambda_j^2)$, $\lambda_j \in [0,1]$

$$p(B_j \mid \lambda_j, c, \varepsilon) \sim \lambda_j N(0, c^2) + (1-\lambda_j) N(0, \varepsilon^2)$$

"Global-local shrinkage priors"

$\lambda_j \sim \text{Bern}(p)$, $\quad \varepsilon \ll c$



Horseshoe

$$B_j \sim N(0, \sigma^2 \tau^2 \lambda_j^2)$$

Global local

$$K_j = \frac{1}{1 + n s_j^2 \tau^2 \lambda_j}$$

$$B_{pm,j} = (1 - K_j)\hat{\beta}_{OLS,j}$$

As $\tau^2 \to 0$

$\hat{B}_{pm} \to 0$

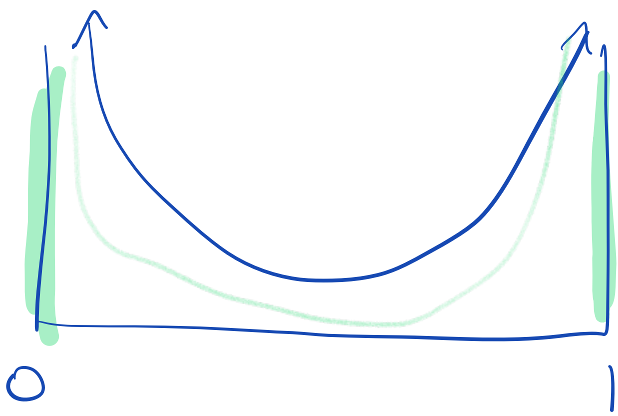
$\tau^2 \to \infty$

$B \to \hat{\beta}_{MLE}$

If $\lambda_j \overset{iid}{\sim}$ Cauchy$^+$ (half-Cauchy)

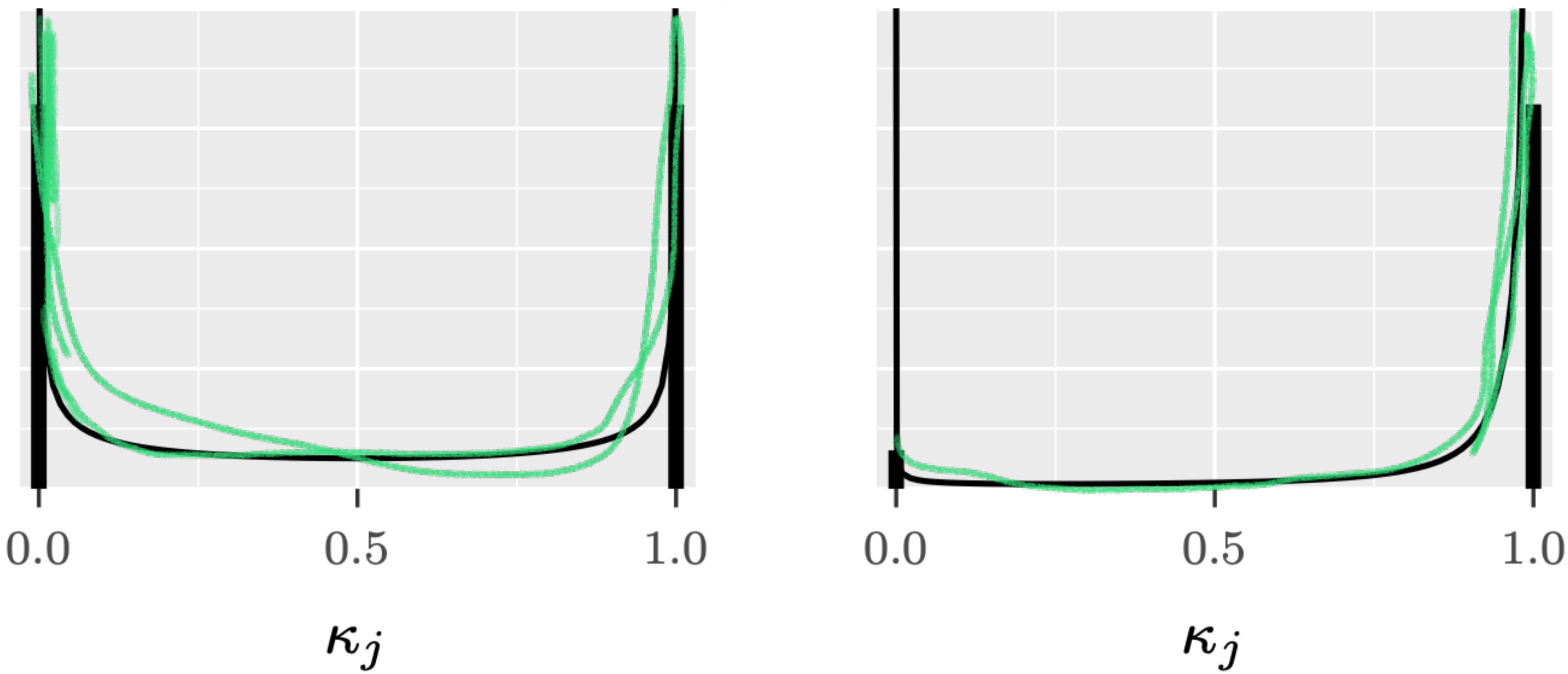
$$\Longrightarrow \quad p(\kappa_j \mid \tau^2) = \frac{1}{\pi} \frac{\tau\sqrt{n}s_j}{(\ )+1} \frac{1}{\sqrt{\kappa_j}\sqrt{1-\kappa_j}}$$

$$\longrightarrow \quad \mathrm{Beta}(1/2, 1/2)$$



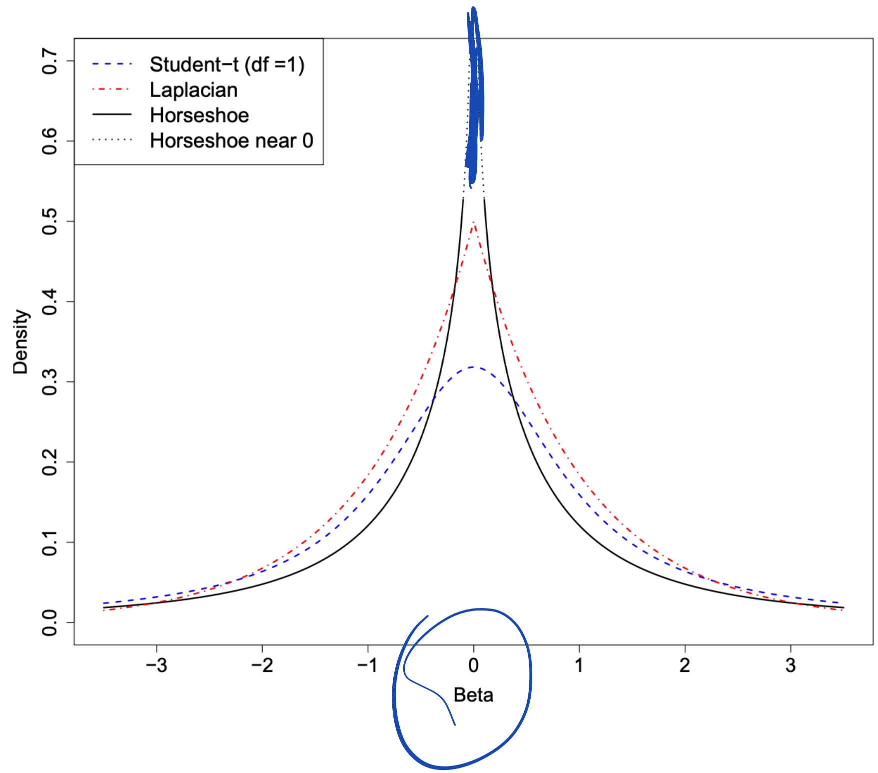
Horsehoe

$\kappa_j = 1$ means shrink all toward 0.



Compring Shrinkage Priors

P(B)



Shrinkage Plots

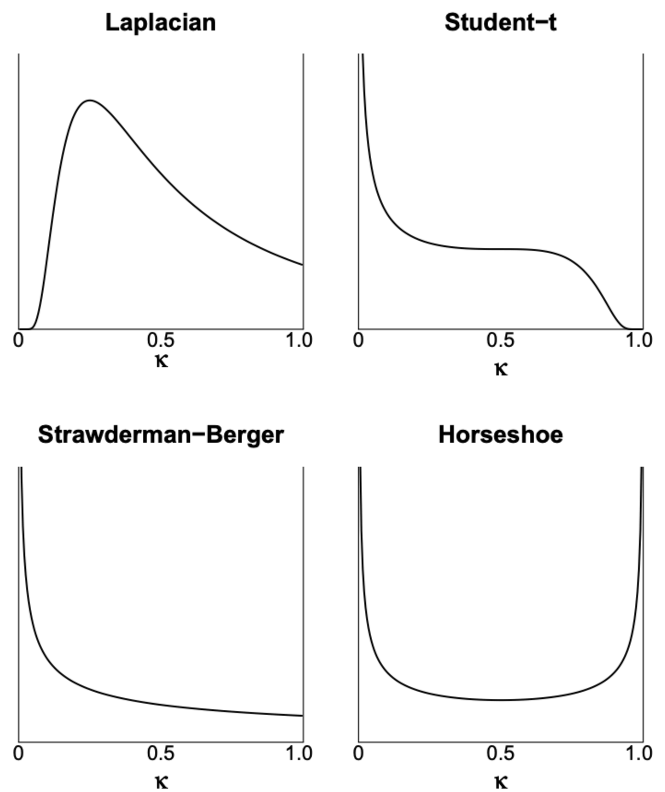


Figure 2: Densities for the shrinkage weights $\kappa_i \in [0, 1]$. $\kappa_i = 0$ means no shrinkage and $\kappa_i = 1$ means total shrinkage to zero.

Finnish Horseshoe

- Original horseshoe favors 0 or $\hat{\beta}$ as solutions
- When coefficients are poorly identified, it's helpful to add additional regulation for the largest coefficients
- The *regularized* horseshoe shrinks all coefficients by at least a small amount

Sparsity information and regularization in the horseshoe and other shrinkage priors

Regularized Horseshoe

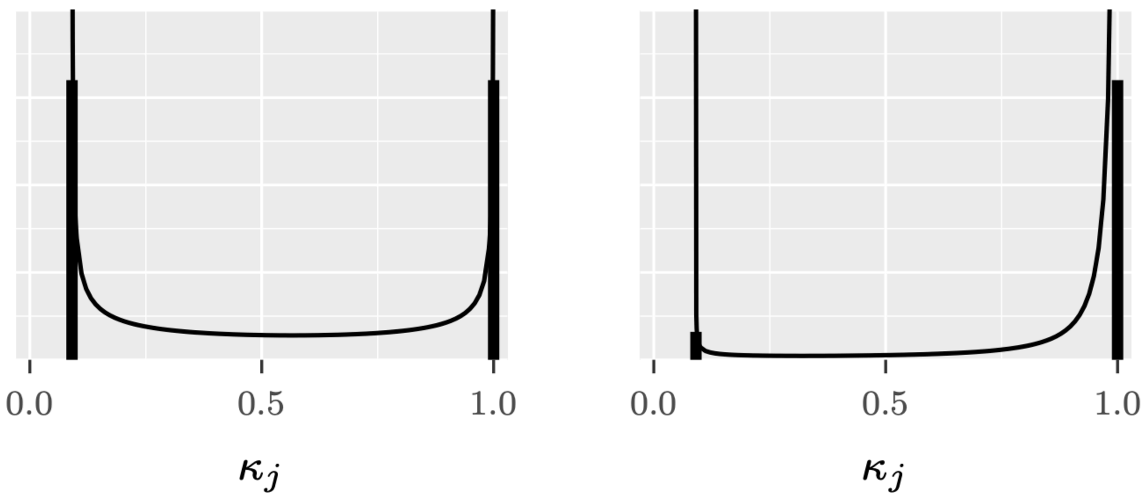


Fig 2: Shrinkage profiles as in Figure 1 but now for the regularized horseshoe (2.8) and spike-and-slab (2.7) with a finite slab width $c = 1$. In comparison to Figure 1, for both priors the first mode is shifted from $\kappa_j = 0$ to $\kappa_j = 1/(1 + n\sigma^{-2}s_j^2c^2)$ (for the plots we have selected $n\sigma^{-2}s_j^2 = 10$).

Global shrinkage parameter

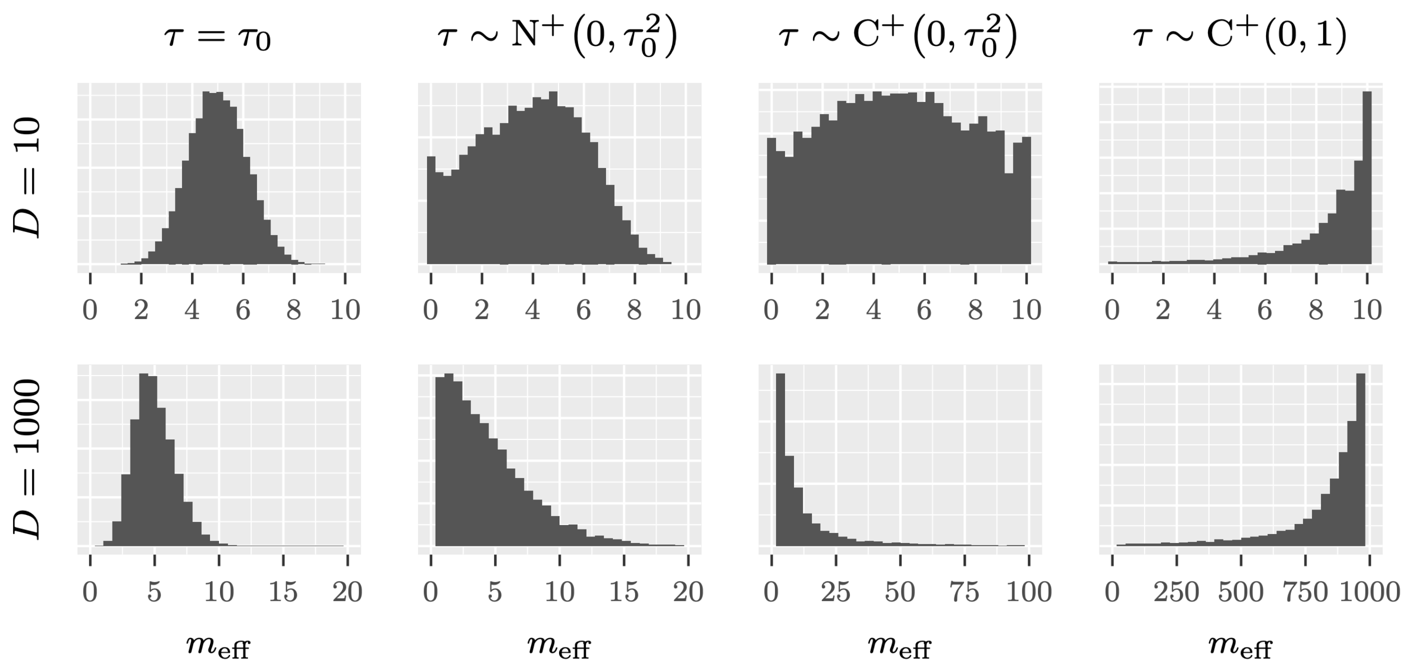


Fig 3: Histograms of prior draws for m_{eff} (effective number of nonzero regression coefficients, Eq. (3.7)) imposed by different prior choices for τ, when the total number of input variables is $D = 10$ and $D = 1000$. τ_0 is computed from formula (3.12) assuming $n = 100$ observations with $\sigma = 1$ and $p_0 = 5$ as the prior guess for the number of relevant variables. Note the varying scales on the horizontal axes in the bottom row plots.

R2-D2 Prior

- R2-induced Dirichlet decomposition (R2-D2)
- A typical approach: specify a p-dimensional prior on all regression coefficients
- Alternative: specify a prior on R^2 and then conditionally on β
  - For a fixed R^2, β lies on an ellipsoid

Bayesian Regression Using a Prior on the Model Fit: The R2-D2 Shrinkage Prior

R2-D2

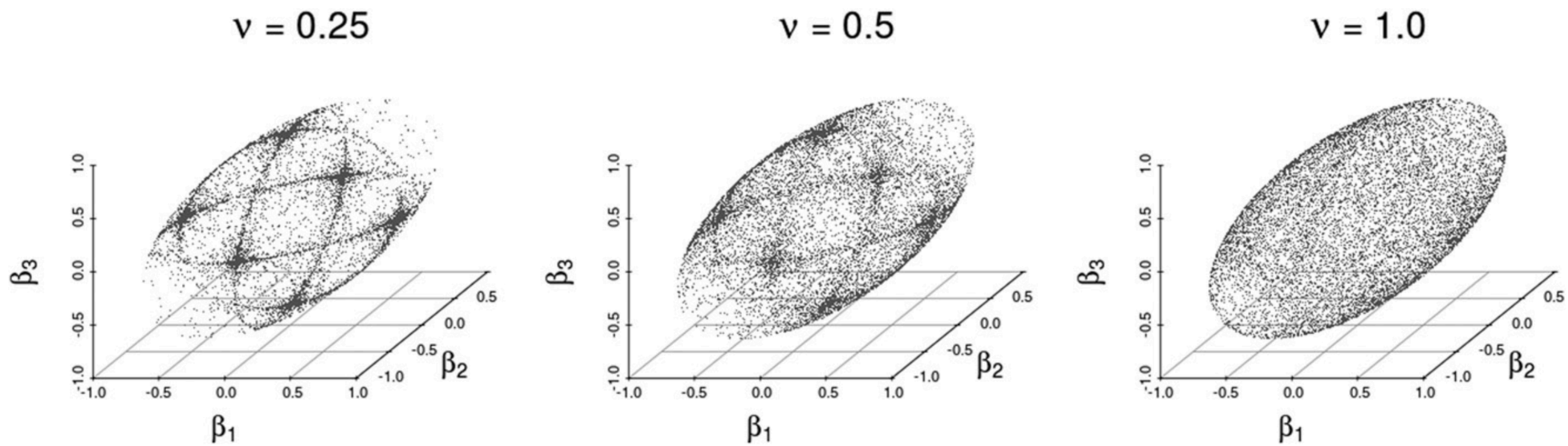


Figure 1. 10,000 samples from the prior of $\boldsymbol{\beta} \mid (R^2, \sigma^2)$ for different choices of ν.

Simulation

- $n = 50, p = 100$ $p > n$
- All coefficients are zero except for three of them
- Coefficients 2,4, and 6 have value 2
- Compare Gaussian, Laplace and Horseshoe priors using rstanarm.

```
library(rstanarm)
n <- 50
p <- 100
beta <- rep(0, p)
beta[1:3] <- 2
x <- matrix(rnorm(n*p), nrow=n)
x <- as.matrix(apply(x, 2, scale))
y <- rnorm(n=n, mean=x %*% beta, sd=1)

p0 <- 10
```

```
tau0 <- p0/(p-p0) * 1/sqrt(n)
fit_hs <- stan_glm(y~x, gaussian(), prior=hs(global_scale=tau0, slab_scale=

posterior <- as.matrix(fit_hs)
hs_dens <- mcmc_areas(posterior, pars=paste0("x", 1:10)) + xlim(c(-3, 3)) +
hs_dens
```

Gaussian Laplace fits

```
fit_gaus <- stan_glm(y~x, gaussian(), prior=normal(scale=1), data=tibble(x=
posterior_gaus <- as.matrix(fit_gaus)
gaus_dens <- mcmc_areas(posterior_gaus, pars=paste0("x", 1:10)) + xlim(c(-5

fit_lap <- stan_glm(y~x, gaussian(), prior=laplace(scale=1), data=tibble(x=
posterior_lap <- as.matrix(fit_lap)
lap_dens <- mcmc_areas(posterior_lap, pars=paste0("x", 1:10)) + xlim(c(-5,
```

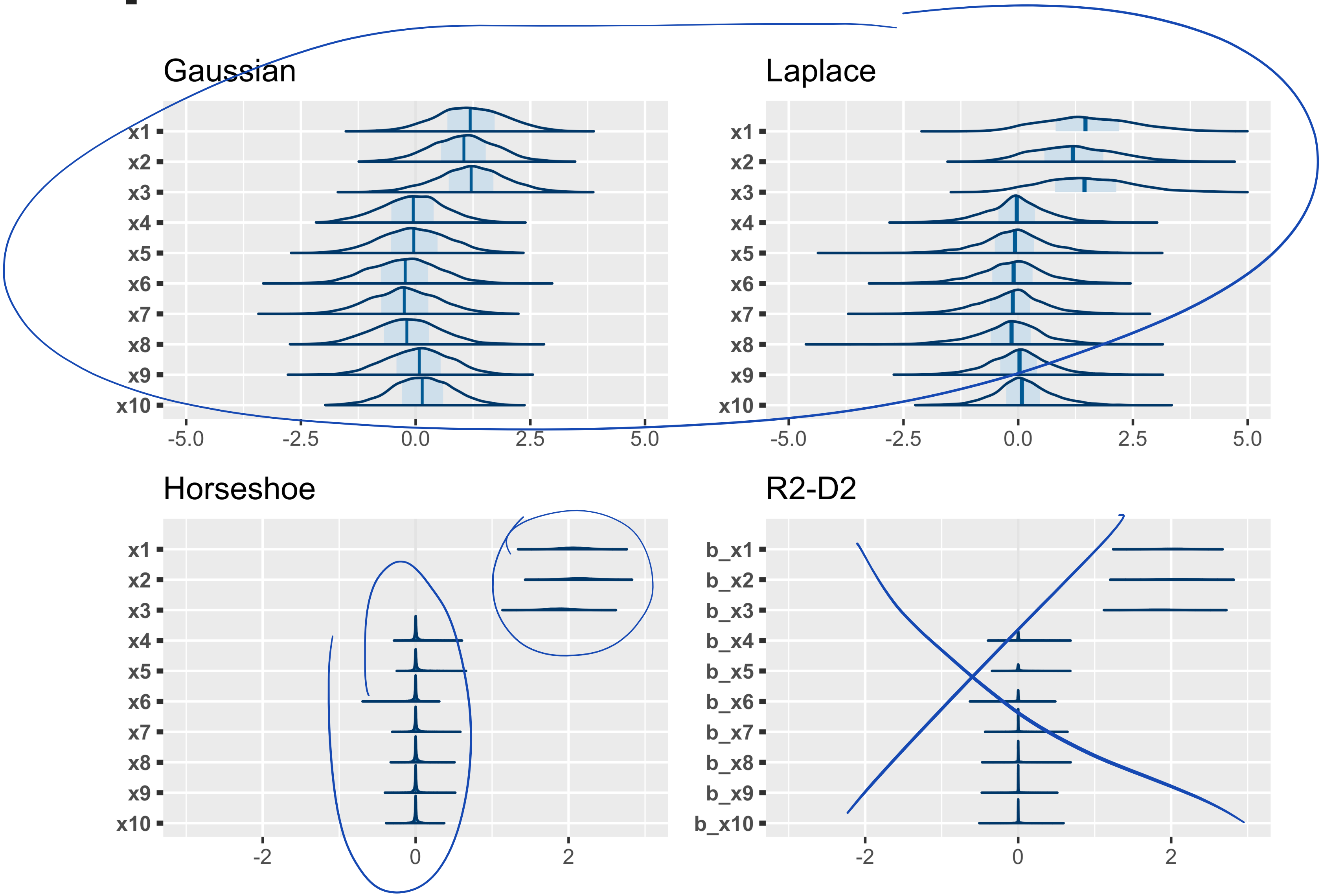
R2-D2 Fit

```r
library(brms)
options(mc.cores=4)
brm_fit <- brm(y ~ x, data=tibble(y=y, x=x),
               prior=set_prior(R2D2(mean_R2 = 0.5, prec_R2 = 2,
                                    cons_D2 = 0.1)),
               family=gaussian())
```

```
Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: ‘Apple clang version 14.0.3 (clang-1403.0.22.14.1)’
using SDK: ‘’
clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -
DNDEBUG   -I"/Library/Frameworks/R.framework/Versions/4.3-
arm64/Resources/library/Rcpp/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.3-
arm64/Resources/library/RcppEigen/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.3-
arm64/Resources/library/RcppEigen/include/unsupported"  -
I"/Library/Frameworks/R.framework/Versions/4.3-
arm64/Resources/library/BH/include" -
I"/Library/Frameworks/R.framework/Versions/4.3-
arm64/Resources/library/StanHeaders/include/src/"  -
I"/Library/Frameworks/R.framework/Versions/4.3-
```

```r
r2d2_dens <- mcmc_plot(brm_fit, type="areas", variable = paste0("b_x", 1:10
```

Comparison Plots



Loo Comparison

```
loo_hs <- loo(fit_hs)
loo_gaus <- loo(fit_gaus)
loo_lap <- loo(fit_lap)
loo_r2d2 <- loo(brm_fit)

loo_compare(loo_hs, loo_gaus, loo_lap, loo_r2d2)
```

```
         elpd_diff se_diff
fit_gaus   0.0       0.0  
fit_hs    -1.2       4.4  
brm_fit   -3.6       3.3  
fit_lap  -10.3       0.9  
```